"Cursed Dungeon" — Procedural Roguelike with Triple AI Mechanics

A compact pitch for a final project in "AI for Games Development": a procedurally generated dungeon crawler where every run is unique, enemies act as believable agents, and the boss employs tactical state-driven behaviour.



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Core Premise — What You Play

You are a lone knight descending into a cursed dungeon. Objectives: survive waves of skeletal enemies, locate and defeat the orc boss, then reach the exit. Each run generates a distinct level layout and encounter rhythm — no two playthroughs are the same.



Combat Encounters

Dynamic skirmishes scale with dungeon topology and spawn distribution.



Boss Encounter

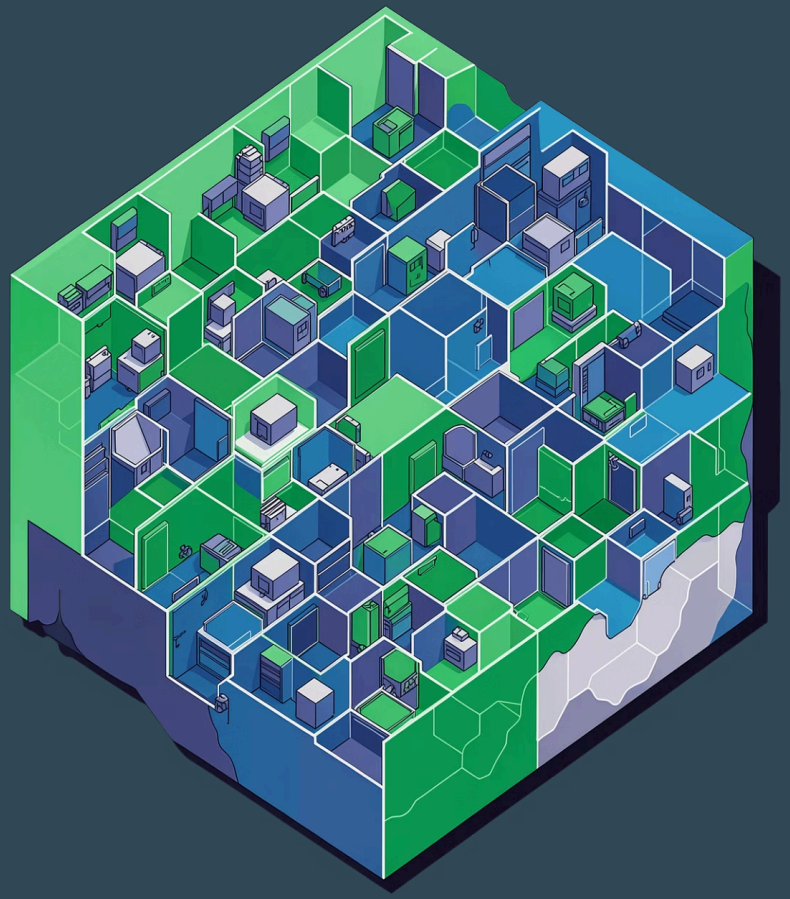
Boss uses stateful AI to adapt tactics mid-fight; fights feel strategic, not repetitive.



Risk & Reward

Players must balance aggression and survival to reach the exit alive.

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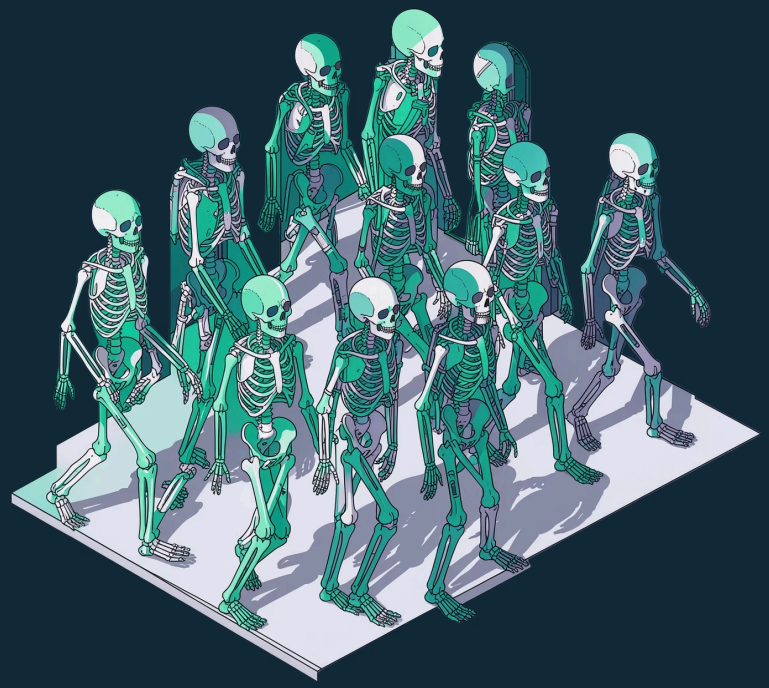
Mechanic 1 — Cellular Automata Level Generation

We use a cellular automaton pipeline to generate organic cave systems that feel handcrafted:

- Initial random fill: each tile seeded as wall or floor.
- Five smoothing passes: tiles adopt wall/floor based on majority of neighbours for natural shapes.
- Post-process BFS connectivity: place player, boss and exit maximally separated to force exploration.
- Optional carve steps: add choke points and rooms to create tactical variety.

Outcome: varied cave topology with emergent chokepoints and flow for combat and navigation.

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Mechanic 2 — Steering Behaviours for Skeleton AI

Skeletons blend three real-time steering forces to produce believable group motion:

Result: emergent encirclement behaviours, dynamic flanking and believable crowding without heavy pathfinding per agent.

Pursue

Direct pursuit of the player when within detection radius; enables targeted pressure.

Separation

Repulsive force to avoid overcrowding; keeps collisions realistic and prevents stack-clumping.

Cohesion

Weak attraction toward local group centre so skeletons move as a swarm rather than scatter.

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Mechanic 3 — Boss AI: Finite State Machine (FSM)

The orc boss is modelled as an FSM with four states and deterministic transitions for transparent, tunable tactics:

1

Patrol

Traverses defined routes in the room; transitions to Chase on sight.

2

Chase

Pursues the player aggressively; if within melee range → Attack.

3

Attack

Stops to perform attack animations and combos; can interrupt to Retreat if HP threshold hit.

4

Retreat

Flees to nearest campfire when HP < 30%, heals over time, then re-enters Chase at full HP.

This FSM creates a boss that reads the battlefield, punishes reckless play, and forces player adaptation.



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Suggested visual for presentation: a clear circular diagram with arrows showing transitions. Use this slide during the FSM explanation to make state logic instantly understandable to instructors.

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Gameplay Loop & Progression

Concise flow that supports replayability and skills growth:

1. **Start:** procedural level generated and spawn points seeded.
2. **Engage:** navigate caves, manage resources, fight skeleton swarms using crowd-aware tactics.
3. **Climax:** boss encounter where FSM dictates tactical phases and recovery windows.
4. **Resolution:** reach exit to win, or reset on death — meta-progression rewards persistent unlocks.

UI elements: player health bar (green → yellow → red), boss health bar, and a kill counter for pacing feedback.

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